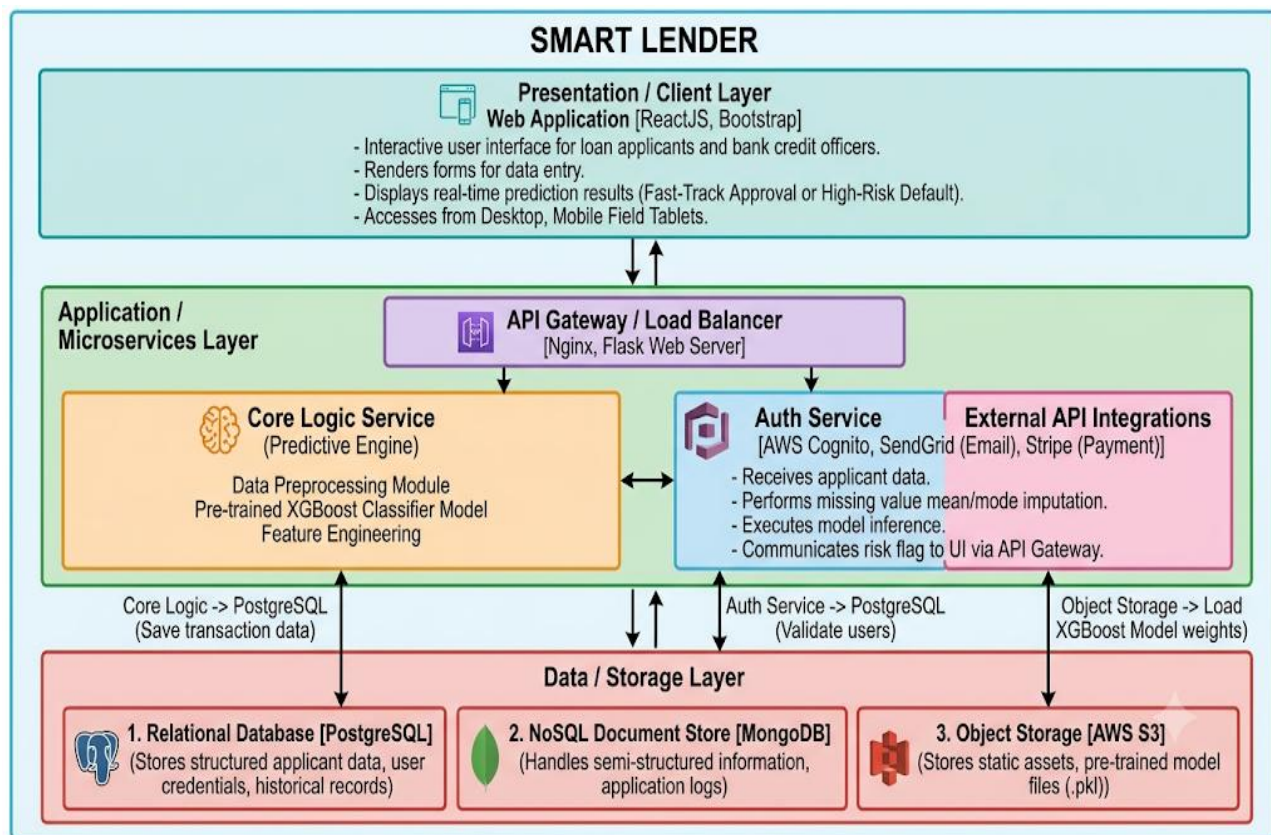


Solution Architecture

Date	8 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Renders an interactive, user-friendly dashboard environment where credit officers, analysts, and loan applicants can seamlessly submit customer profile metrics and instantly view visual prediction eligibility outputs.	HTML5, CSS3, Bootstrap
[API Gateway]	Functions as the central request routing layer that captures multi-attribute web form inputs from the UI, manages client connection endpoints, and maps them to the server-side predictive application routines.	Flask Web Routing / Built-in Werkzeug Server
[Microservices]	Contains the intelligent computational core of the system. This layer handles missing data preprocessing (mean and mode imputation), encodes variables, and runs real-time automated loan approval or high-risk default classification inference using a pre-trained ensemble model.	Python, Scikit-Learn, XG Boost Classifier Engine
[Database]	Operates as the storage and persistence layer that loads serialized predictive model configurations into live memory arrays and temporarily references the historical tabular structures used for context pipeline alignment.	Serialized Pickle (.pkl), Pandas DataFrames